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Understanding the mechanisms behind micro-aeration to enhance anaerobic digestion of corn straw

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ABSTRACT

Micro-aeration has been shown to be favorable for anaerobic digestion (AD) process. However, the underlying mechanisms are not fully revealed. In this study, the effects of micro-aeration on AD process were investigated, the underlying mechanisms were also explored by comprehensively analysis of the substrate degradation, intermediate products, electron-transfer capacity, key enzymes and microbial communities. Results showed micro-aeration could benefit the AD process, and the beneficial effects were oxygen intensity-dependent. Maximum methane yield (262.0 ± 6.0 mL/(g·VS)) was achieved at micro-aeration of 0.2 mL/(g·VS·day), which was 7.8% higher than that of the control (without micro-aeration). The volatile fatty acids concentration, cellulase activity, soluble humic acids (SHAs) concentration and substrate degradation rate were all enhanced under micro-aeration condition. Moreover, micro-aeration could regulate the bacterial and archaeal communities: bacterial diversities and relative abundances of facultative and acetate-oxidizing bacteria (e.g., *Firmicutes, Synergistota* and *Spirochaetota*) increased, leading to higher substrate utilization rate; archaeal diversities decreased while oxygen-tolerating methanogens (e.g., *Methanobacterium*) became dominant to promote the methanogenesis. AD system with micro-aeration showed higher electron-transfer capacity and better conductive environment due to the increased SHAs concentration. The increased relative abundance of direct-interspecies-electron-transfer (DIET)-associated microbes could be favorable to DIET-based syntrophic methanogenesis which promotes AD under micro-aeration.

1. Introduction

Lignocellulosic biomass, such as agricultural and forest residues, energy crops, and grasses is the most abundant renewable bioresource on the Earth [1]. Corn is one of the most important grain crops in China, and the yearly yield of corn crop is estimated to be 520 million tons, which yields 260 million tons of corn straw as a major by-product [2,3]. Corn straw has been proved to be a potential substrate for biogas production via anaerobic digestion (AD). Many biogas plants that use corn straw as feedstock are built every year in China. Although promising, some barriers still exist in AD of corn straw, among which the low hydrolysis efficiency caused by the recalcitrant structure of corn straw is generally regarded as the rate-limiting step [4].

In order to enhance the AD efficiency of corn straw, various substrate pretreatment methods have been extensively studied, including physical, chemical and biological pretreatments as well as their combinations [5–8]. However, these pretreatment methods have their respective inherent shortcomings, e.g., need of chemicals, high energy consumption and/or long residence time. Another choice to improve the AD efficiency of cellulosic substrate is to manage or alter the microbial community composition through bioaugmentation, enrichment and/or stimulation of specific indigenous microorganisms [9]. Micro-aeration by introducing limited amount of oxygen into the AD system has been proved to be a potential strategy to address the challenge of low hydrolysis efficiency associated with AD of cellulosic substrate [10,11]. As reported, micro-aeration during AD can stimulate the growth of facultative anaerobes which excrete more extracellular enzymes to improve hydrolysis and fermentation of lignocellulosic biomass, thus enhancing

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the AD efficiency [12–14]. Given its potential economic and environmental benefits, micro-aeration during AD has received increasing attention. However, no consistent even contradictory conclusions have been made about the effect of micro-aeration on the AD process. Some researchers reported that micro-aeration increased the hydrolysis efficiency, VFAs (volatile fatty acids) formation and methane yield during AD [15–17], while others found no even negative effect of micro-aeration on the AD process [18–20]. Such a conflict could be because the underlying mechanisms of micro-aeration remain unclear. Furthermore, the responses of bacteria and archaea, the enzyme activities and the degradation characteristics of different components (e.g., cellulose, hemicellulose and lignin) under micro-aeration conditions were rarely reported.

Therefore, this work aimed to investigate the effects of micro-aeration on AD of corn straw. In addition, the underlying mechanisms of how micro-aeration benefit the AD process were explored using comprehensive analytic methods include intermediate products analysis, model-based analysis, microbial community structure analysis, cyclic voltammetry (CV) and conductivity tests.

2. Materials and methods

2.1. Substrate and inoculum

Corn straw was collected from a corn farm in Xuzhou, Jiangsu Province, China and used as a substrate in this study. The corn straw was air-dried to the moisture content of 6.6 ± 0.2%, and ground to 0.5-cm particles before used. The inoculum used in this study was obtained from an AD reactor which uses corn straw as a substrate and is operated at 37 °C. Before used, the inoculum was filtered through a 10-mesh sieve to remove large particles (e.g., undigested corn straw). The characteristics of substrate and inoculum were shown in Table 1.

2.2. Batch biomethane potential (BMP) test

The batch mesophilic (37 °C) AD of corn straw was carried out in triplicate with a total fermentation time of 40 days. Before AD, 5.6 g (TS) corn straw and 2.0 g (TS) inoculum were mixed in 300-mL glass bottles. The biogas production with the inoculum only (i.e., no corn straw) was also carried out as a blank group. The working volume was adjusted to 200 mL by adding nutrient solution as described by Angelidaki et al [21]. All the bottles were then purged with argon for 5 min to remove the air and sealed with rubber stoppers. All the bottles were incubated in a shaking air bath at 37 °C with the shaking speed of 200 rpm. In order to create different micro-aeration intensity, oxygen of different doses was injected into the bottles daily by using a syringe. The tested oxygen doses included 0 (control), 0.05, 0.1, 0.2, 0.4, 0.8 and 1.6 mL/(g·VS·day), which were denoted by MA0, MA0.05, MA0.1, MA0.2, MA0.4, MA0.8 and MA1.6, respectively.

2.3. Model-based analysis

The modified Gompertz model (Eq. (1)) [22] was adapted to fit the methane production data for analyzing the effects of micro-aeration on AD of corn straw.

$$M_t = M_m \times exp\left\{ -exp\left[\frac{R_m \times e}{M_m}(\lambda - t) + 1\right]\right\} \quad (1)$$

where, M_t represents the cumulative methane production at time t (mL/g·VS), M_m represents the methane yield potential (mL/g·VS), R_m represents the maximum methane yield rate (mL/(g·VS)·d), λ represents the lag time (d), t represents the digestion time (d), and $e = 2.73$.

Table 1
Characteristics of substrate and inoculum.

Characteristics	Substrate (corn straw)	Inoculum
Moisture content (%)	6.6 ± 0.2	97.6 ± 0.1
Total solid (TS, %)	93.4 ± 0.2	2.4 ± 0.1
Volatile solid (VS, % of TS)	90.7 ± 0.2	68.2 ± 0.2
C (% of TS)	36.7 ± 1.6	–
N (% of TS)	1.3 ± 0.4	–
H (% of TS)	4.9 ± 0.4	–
C/N	28.2 ± 1.2	–

Note: "–" means this parameter was not tested.

2.4. Microbial community structure analysis

When the operation of AD systems became stable (i.e., the 13th day of AD in current study), 10 mL of digestate were collected from three representative AD groups such as MA0 (control without micro-aeration), MA0.2 (positive effect of micro-aeration) and MA1.6 (overdose of micro-aeration) for microbial community structure analysis. The microbial community structure was analyzed by the Majorbio Bio-Pharm Technology (Shanghai, China) using the high throughput sequencing method. According to the instructions, the E.Z.N.A.® DNA Kit (Omega Biotek, Norcross, GA, USA) was used to extract the DNA from the digestates. In order to run the polymerase chain reaction (PCR), a primer set of 338F (ACTCCTACGGGAGGCAGCAG) and 806R (GGACTACHVGGGTWTCTAAT) was used to amplify the V3-V4 region of the bacterial 16S rRNA gene, and a primer set of 524F (TGYCAGCCGCCGCGGTAA) and 958R (YCCGGCGTTGAVTCCAATT) was used to amplify the V4-V5 region of the archaeal 16S rRNA gene [23]. The quantification of the PCR products was performed with the QuantiFluor™-ST blue fluorescence quantitative system (Promega Inc., China). The Illumina MiSeq platform was used to pool the obtained amplicons from the PCR reactions in equimolar and paired-end sequenced.

The QIIME package was used to extract the high-quality sequences (Quantitative Insights Into Microbial Ecology) (V1.9.1). Raw data were first checked for barcode and adapter integrity, and the low-quality sequences that contained ambiguity and less than 200 basis points were deleted to obtain effective sequencing data. The operational taxonomic unit (OTU) for the sequence set was classified by RDP Classifier (RibosomalDatabase Project) (V2.11) with the consistency threshold of 97% identity. The identification and removal of chimeric sequences were done by using Usearch (V7.1), and the classification of SILVA132 16S rRNA database was done by using RDP Classifier with the confidence threshold of 80%.

2.5. Analytical methods

2.5.1. Excitation emission matrix (EEM) fluorescence spectra for digestate analysis

EEM fluorescence spectra was applied to characterize dissolved organic matter in the digestates to study the effect of oxygen dose on the dissolved organic matter. After 40 days of fermentation, 1 mL digestate was filtered through 0.45-μm filters and diluted 5x using deionized water before the EEM measurement by a fluorescence spectrophotometer (F-7000 FL, HitachiCo, Japan). The excitation wavelength was gradually increased from 200 to 450 nm while emission wavelength was gradually increased from 250 to 650 nm by a 5-nm increment. The three-dimensional fluorescence region formed by the excitation and emission wavelengths was used to divide the organics in each digestate into five groups as shown in Table 2.

2.5.2. Analysis of soluble humic acids (SHAs) in the digestate

According to the polyphenol theory of humus formation, lignin, cellulose and other non-lignin substances can be converted to humic acids (HAs) or fulvic acids by microbes in AD process (**Fig. S1**) [25]. Thus, high concentration of HAs or fulvic acids in the digestates could indicate high degradation of corn straw. Moreover, SHAs were also

Table 2
Organics grouped by three-dimensional fluorescence region [24]

Group	EX (nm)	EM (nm)	Organics
I	200–250	250–330	Tyrosine-like
II	200–250	330–380	Tryptophan-like
III	200–250	380–650	Fulvic acid-like
IV	250–340	250–380	Soluble microbial byproduct-like
V	250–450	380–650	Humic acid-like

reported to serve as electron transport media and benefit the direct interspecies electron transfer (DIET)-based syntrophic methane production [24,26]. SHAs in the digestate were extracted according to the procedure developed by the International Humic Substances Society (IHSS) [27]. Before the extraction of SHAs, 20 mL samples were centrifuged at 5000 rpm for 15 min and then filtered through 0.45-um filters. To the filtrates, 6 mol/L HCl was added to reduce the pH to 1.0 for precipitating out HAs, and 6 mol/L NaOH was then used to dissolve the HAs precipitates. Such a procedure of precipitation-dissolution-precipitation was repeated for three times to purify the SHAs precipitates. The SHAs were quantified in g/L according to the gravimetric method [28]. Fourier transform infrared (FTIR) spectroscopy of SHAs precipitates (freeze-drying) were conducted by using IRTracer-100, Shimadzu, Japan.

2.5.3. Lignocellulosic compositions analysis of corn straw

Extractives of corn straw were first removed by water and ethyl alcohol extraction for 24 h [29], and the residual extractive-free corn straw was used for the analysis of structural carbohydrates and lignin [30]. Briefly, the extractive–free corn straw underwent a two-step acid hydrolysis: 72% sulfuric acid (3.00 ± 0.01 mL) was first used to hydrolyze corn straw (300.0 ± 10 mg) in a water bath at 30 °C for 60 min, and then the acid was diluted to 4% by adding 84.00 ± 0.04 mL of deionized water and autoclaved for 60 min at 121 °C. The hydrolysate was vacuum-filtered through a filtering crucible. The acid insoluble lignin of corn straw was measured by weight difference before and after the ashing of the solid residue, and the acid soluble lignin in the filtrate was determined on a UV–Vis spectroscopy (P1, MAPADA Co, China) at 320 nm. The monosaccharides in the filtrate were quantified by using a high pressure ion chromatography (DIONEX ICS-5000 + SP-5, Thermo Fisher Scientific, USA) equipped with CarboPac PA20 column (3-mm diameter × 150-mm length). The injection volume was 5 μL, and the mobile phase consisted of 1 mol/L sodium acetate and 250 mmol/L NaOH in HPLC grade water.

2.5.4. Cyclic voltammetry (CV) and conductivity test of digestate

CV analysis could be performed to illustrate the redox property of special material and/or system [31]. The digestate samples from MA0, MA0.2 and MA1.6 were firstly centrifuged at 8000 rpm for 10 min, the supernatants were used for CV and conductivity test. The CV experiment was carried out to excavate the redox reactions in digestate with a three-electrode electrochemical cell in a single chamber [32]. The reference electrode, counter electrode and working electrode in the electrochemical cell were saturated calomel electrode (SCE, Huachen, China), Pt and carbon cloth (2*2 cm), respectively. Digestate of 140 mL was added into the electrochemical cell, and the cell was purged with N_2 for 5 min to remove the dissolved oxygen. CV was tested on an electrochemical workstation (CHI600E, Huachen, China) with a scanning voltage in the range of −1.0 ~ 1.0 V (versus SCE) at 50 mV/S scan rate. The electric conductivity shows the talent of electron transfer for specific material and/or system, which could their abilities in participation of DIET [33]. The conductivity of digestate was measured by using conductivity meter (DDS-307, Leici, China).

2.5.5. Cellulase activity analysis

The cellulase activity was determined by using DNS (dinitrosalicylic acid) method [34]. Samples were first centrifuged at 11,800 rpm for 15 min at 4 °C, and the supernatants were collected and balanced at 37 °C for 10 min. The mixture of 1.00 mL of sodium carboxymethyl cellulose (CMC-Na), 2.00 mL of DNS, 0.50 mL of balanced sample, and 5 mL of deionized water was incubated in a water bath at 37 °C for 60 min. Afterwards, the mixture was heated in boiling water for 5 min to stop the enzymatic reaction and cooled down to ambient temperature. The mixture was then diluted to 25 mL, and the absorbance was read at 540 nm on a UV–Visible spectrophotometer (P1, MAPADA Co, China). The calibration curve was made by using glucose as a standard with a series of concentrations.

2.5.6. Other analyses

The water displacement method was used to monitor the biogas production in volume, and the methane content of biogas was analyzed by using a gas chromatograph equipped with a thermal conductivity detector and a Porapak Q stainless steel analytical column (OD × L = 3 mm × 180 cm) (SP 6890, Shandong Lunan Inc., China) [35]. The pH was determined using a compact pH-meter (LAQUAtwin-pH-33, Horiba Co., Japan). The TS and VS were determined according to the gravimetric method. To measure VFAs concentration, samples were first centrifuged at 5000 rpm for 15 min, and the supernatants were filtered through 0.45-μm filters. After acidified with 3 M phosphoric acid, the VFAs concentration was analyzed on a gas chromatograph (GC-2010, Shimadzu, Japan). The total VFAs were determined by summing up each individual VFA (i.e., acetate, propionate, butyrate/isobutyrate, valerate/ isovalerate, caproate, heptanoate and caprylate). All analyses were conducted in triplicates unless specified, otherwise. The one-way ANOVA (analysis of variance) was performed to test the significance of difference between different treatments with $P_{critical} = 0.05$ using SPSS software (ver. 16.0, IBM, USA).

3. Results

3.1. Methane production from anaerobic digestion of corn straw

As shown in Fig. 1, the AD reactors treated by different micro-aeration intensities had similar trends in cumulative methane yield, while the amount of cumulative methane yield was significantly affected by the micro-aeration ($p < 0.05$). The highest 40-day cumulative methane yield of 262.0 ± 6.0 mL/(g·VS) was obtained with micro-aeration intensity of 0.2 mL/(g·VS·day) (MA0.2), which was 7.8%

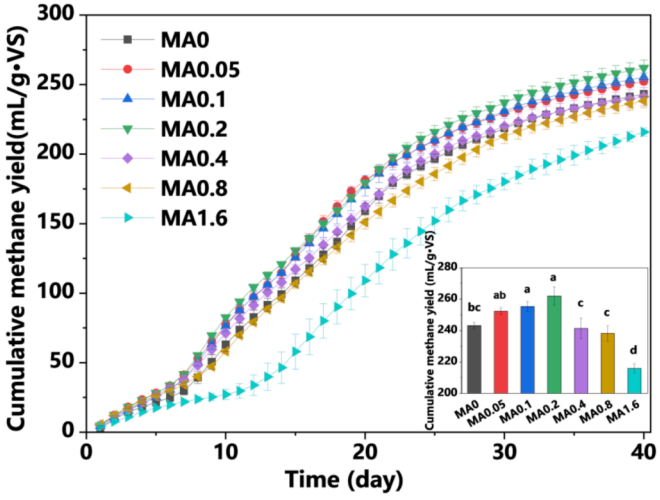


Fig. 1. Cumulative methane yield of anaerobic digestion of corn straw under different micro-aeration intensity. Error bars represent standard deviations of triplicate tests. The methane yield has been corrected by subtracting the methane yield of the blank group (i.e., the AD with inoculum only).

higher than that of MA0. However, further increase of micro-aeration intensity impaired the methane yield compared to MA0. When micro-aeration intensity reached 1.6 mL/(g·VS·day) (MA1.6), the cumulative methane yield was only 215.9 ± 2.9 mL/(g·VS), which was 11.2% lower than that of MA0. In current study, the whole AD process was divided into three stages based on the fermentation time, including early AD stage (the 1st ~ 13th day), middle AD stage (the 14th ~ 26th day) and late AD stage (the 27th ~ 40th day) [36]. The cumulative methane yields of different AD stages were shown in Table 3. Compared to MA0, the methane production increased with the increase of micro-aeration intensity from 0.05 to 0.2 mL/(g·VS·day). In the early AD stage, the highest methane yield of 113.0 ± 2.4 mL/(g·VS) was obtained from MA0.2, which was 23.3% higher than that of MA0. Further increase of micro-aeration intensity reduced the methane yields, i.e., the methane yields of MA0.8 and MA1.6 are only 96% and 43% of MA0, respectively. In the middle AD stage, the methane yield of MA1.6 was 120.1 ± 2.7 mL/(g·VS), which accounted for 55% of its total (40 days) methane yield and was 9%, 10%, 10%, 11%, 15% and 16% higher than that of MA0, MA0.05, MA0.1, MA0.2, MA0.4 and MA1.6, respectively. In the late AD stage, the highest methane production (56.0 ± 5.7 mL/g·VS) was still obtained by MA1.6, which was 35% higher than that of MA0.

In order to further analyze the effect of micro-aeration on AD, the modified Gompertz was used to fit the experimental data of methane production. The fitting results are shown in Table 4. The experimental data were well fitted to the modified Gompertz model, and all the R^2 values are higher than 0.99. The micro-aeration intensity less than or equal to 0.2 mL/(g·VS·day) raised methane yield potential (M_m) and maximum methane production rate (R_m) but reduced lag phase time (λ). Further increase of micro-aeration intensity lowered the methane yield potential and the maximum methane production rate but increased the lag phase time. The micro-aeration also affected the time required to produce 80% of the total methane yield (T_{80}), which increased with the increase of micro-aeration intensity from 0.05 to 1.6 mL/(g·VS·day), and the shortest T_{80} was obtained by MA0.05 (24.5 days), which was 2.3 days shorter than that of MA0.

3.2. Impact of micro-aeration on daily methane yield, cellulase activity and volatile fatty acids

Three groups (MA0, MA0.2 and MA1.6) were selected and compared to analyze the effects of micro-aeration on the AD process. As shown in Fig. 2, the daily methane yields of MA0 and MA0.2 had the same tendency. There were two methane production peaks for MA0 and MA02 and the maximum peak values were obtained at the 10th and 9th day of AD, respectively. However, the daily methane yield of MA1.6 only showed one methane production peak at the 15th day of AD, which might be caused by the inhibition of the methanogen activity under high micro-aeration intensity.

The variations of cellulase activity and total VFA (TVFA) concentrations during AD of corn straw are demonstrated in Fig. 3. The cellulase activity of these three groups showed the same tendencies: raised rapidly at the first 6 days and leveled off afterwards with slight fluctuation until the end of AD (Fig. 3**a**). The cellulase activity of MA0, MA0.2 and MA 1.6 showed no obvious difference during the first 6 days of AD. After that, the cellulase activity of MA0.2 and MA 1.6 reached the peak values at the 12th day, which were 15% and 20% higher than that of MA0, respectively.

According to Fig. 3**b**, the TVFA concentrations under micro-aeration conditions (MA0.2 and MA1.6) were higher than that of MA0 after the 3rd day of AD and stayed at high level for a longer period. The peak value of the TVFA concentration of MA0 was 2648 ± 75 mg/L at the 3rd day, while the TVFA peak concentrations of MA0.2 and MA1.6 were 2368 ± 81 mg/L and 2396 ± 56 mg/L at the 6th and 9th day, respectively. The delay of the TVFA peaks under micro-aeration might be due to the fact that oxygen promotes the growth of facultative bacteria that could consume part of VFAs [11]. However, the VFAs under micro-aeration condition remained much higher concentration for longer time. This could be explained by the fact that the facultative bacteria could excrete more extracellular cellulase to enhance the hydrolysis of corn straw, thus producing high VFAs concentrations (i.e., MA0.2 and MA1.6). Moreover, enzyme activity and VFAs were close but the methane yield differed between MA0.2 and MA1.6. The reason behind this phenomenon might be that methanogenesis was inhibited by excess oxygen and VFAs were consumed by facultative bacteria using oxygen as an electron acceptor in MA1.6.

3.3. Release of soluble humic acids during AD of corn straw

As shown in Fig. 4**a-c**, fluorescence responses were weak in Regions I, II, III and IV, but the fluorescence response in Region V that represents humic acid-like organics was strong and became stronger with the increase of oxygen dosage. The SHAs concentrations of MA0, MA0.2 and MA1.6 were 1.31 ± 0.06 g/L, 1.67 ± 0.12 and 2.05 ± 0.10 g/L, respectively (Fig. 4**d**). These findings suggested that the addition of

Table 3
Cumulative methane yield at different AD stage.

Group	Methane yield (mL/(g·VS))		
	Early AD stage (1st-13th day)	Middle AD stage (14th −26th day)	Late AD stage (27th-40th day)
MA0	91.6 ± 4.7	110.2 ± 1.7	41.5 ± 1.0
MA0.05	105.8 ± 3.5	108.6 ± 0.6	37.9 ± 1.4
MA0.1	106.1 ± 3.9	109.1 ± 2.7	40.2 ± 1.6
MA0.2	113.0 ± 2.4	108.0 ± 3.6	41.0 ± 1.1
MA0.4	100.7 ± 3.2	104.5 ± 2.8	36.1 ± 3.1
MA0.8	88.5 ± 2.4	103.1 ± 2.6	46.6 ± 0.2
MA1.6	39.8 ± 9.9	120.1 ± 2.7	56.0 ± 5.7

Table 4
Regression results of the modified Gompertz model.

Group	Kinetic model parameters				
	M_m (mL/g·VS)	R_m [mL/(g·VS·day)]	λ(d)	T_{80} (d)	R^2
MA0	255.7 ± 1.5	10.5 ± 0.1	4.4 ± 0.1	26.8	0.999
MA0.05	259.3 ± 1.1	11.3 ± 0.1	3.4 ± 0.1	24.5	0.999
MA0.1	263.6 ± 1.1	11.3 ± 0.1	3.6 ± 0.1	25.0	0.999
MA0.2	272.0 ± 1.4	11.2 ± 0.1	3.1 ± 0.1	25.4	0.999
MA0.4	253.8 ± 1.3	10.3 ± 0.1	3.4 ± 0.1	26.1	0.999
MA0.8	256.1 ± 1.2	9.5 ± 0.1	3.8 ± 0.1	28.6	0.999
MA1.6	240.9 ± 5.2	8.9 ± 0.2	7.7 ± 0.3	32.6	0.996

Note: T_{80} means the time required to achieve 80% of the total methane yield.

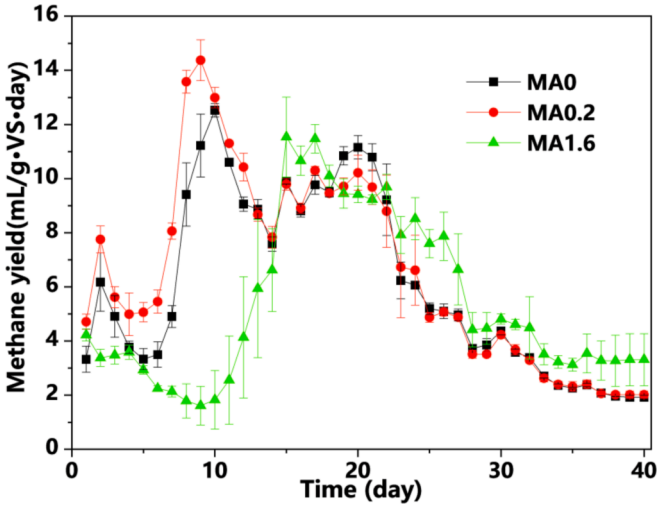


Fig. 2. Daily methane yield of MA0, MA0.2 and MA 1.6.

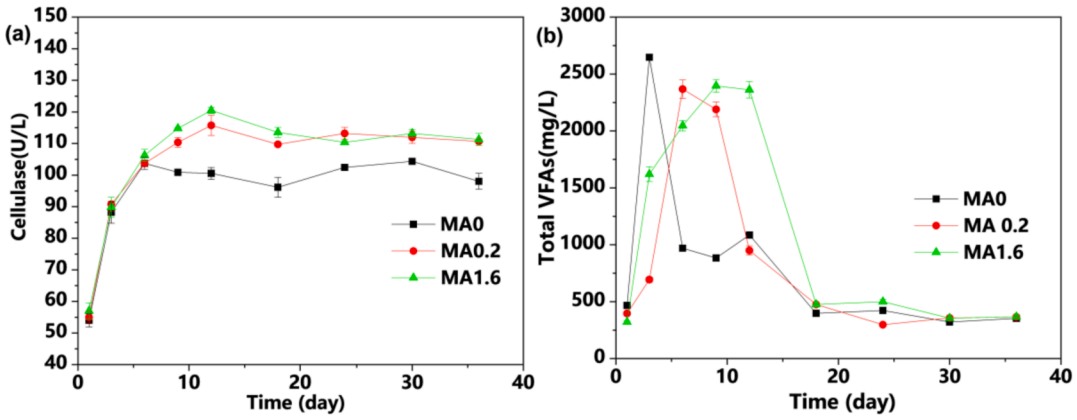


Fig. 3. Cellulase activity (a) and total VFA concentrations (b) during AD of corn straw.

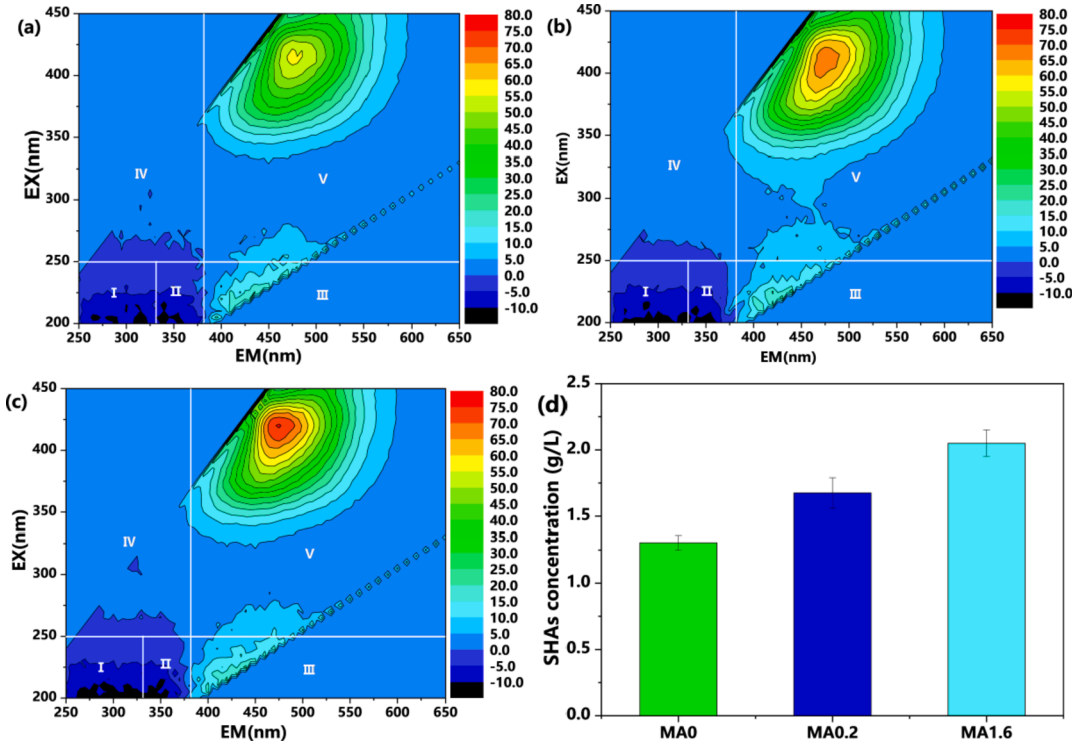


Fig. 4. EEMs images of MA0 (a), MA0.2 (b) and MA1.6 (c) and soluble HAs concentrations in the digestates (d). I: Tyrosine-like, II: Tryptophan-like, III: Fulvic acid-like, IV: Soluble microbial byproduct-like and V: Humic acid-like organics.

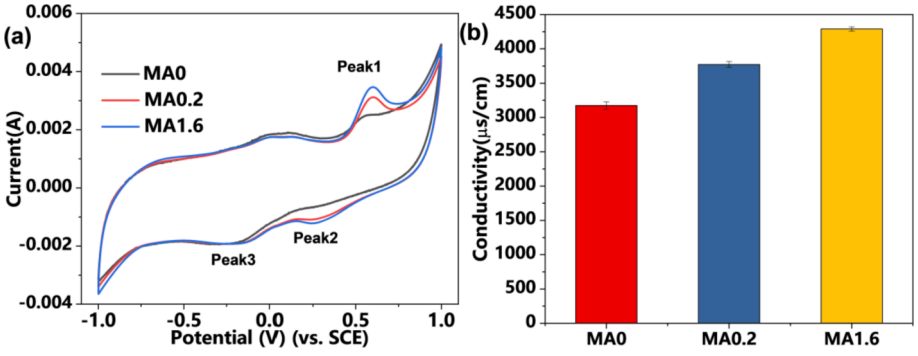


Fig. 5. Cyclic voltammetry (a) and conductivities (b) of digestate from MA0, MA0.2 and MA1.6.

oxygen enhanced the degradation of corn straw and led to increase the SHAs concentration, which could benefit the DIET based syntrophic methane production in turn. The roles of HAs in AD of sludge were intensively studied [24,37]. However, little research has been done on the generation of HAs from AD of corn straw, and the effects of HAs on the AD of lignocellulosic biomass need to be further explored.

3.4. Cyclic voltammetry and conductivity of digestate

Electrochemical methods can be used to characterize these redox reactions, therefore, CV experiment was conducted in this study to analyze the electron transfer capacity of the AD system [38]. Fig. 5**a** shows CVs of MA0, MA0.2 and MA1.6. It could be observed that three distinct redox peaks with potentials of 0.60 (peak1), 0.27 (peak2) and −0.24 V (peak3) (versus saturated calomel electrode) presented in all three groups at the scan rate of 50 mV/s, however, the peak currents of MA0.2 and MA1.6 were different. The peak1 currents of MA0.2 and MA1.6 were 24% and 38% higher than that of MA0, and the peak2 currents of MA0.2 and MA1.6 were 75% and 96% higher than that of MA0, while peak3 currents showed no difference among three groups. The higher peak currents observed in digestate from MA0.2 and MA1.6 demonstrated their higher capacity of electron transfer, which could attribute to the increased SHAs concentrations. FTIR analysis showed that freeze-dried SHAs possessed abundance of surface oxygen-containing functional groups (C—O, C═O and C—O; shown in **Fig. S2**), which could be used as electron shuttle to accept and donate electrons, leading to higher capacity of electron transfer [31]. Conductivity was crucial to triggering the direct interspecies electron transfer (DIET)-based syntrophic methane production in anaerobic digestion [39]. The conductivity values of digestate sampled from MA0, MA0.2 and MA1.6 were 3,173 ± 52, 3,773 ± 43 and 4,290 ± 30 μs/cm, respectively (Fig. 5**b**). The results indicated that micro-aeration could create a better conductive environment for syntrophic microorganisms.

3.5. Variation of structural compositions and component removal of corn straw during AD

The structural compositions of raw corn straw and biogas residue of MA0, MA0.2 and MA1.6 are shown in Fig. 6a. The contents of cellulose, hemicellulose and lignin of substrate (raw corn straw) were 34.00 ± 2.01%, 19.45 ± 4.55% and 22.71% ± 0.71%, respectively. The chemical composition analysis of the biogas residues showed that the cellulose contents of MA0, MA0.2 and MA1.6 decreased to 22.75 ± 2.75%, 21.59% ± 0.59% and 21.59% ± 1.59%, respectively after AD. Similarly, the hemicellulose contents of MA0, MA0.2 and MA1.6 decreased to 10.30% ± 1.69%, 10.46% ± 0.53% and 9.46% ± 1.46%, respectively after AD. In contrast, the lignin contents of MA0, MA0.2 and MA1.6 increased to 33.83% ± 2.17%, 33.22% ± 1.22% and 33.26% ± 3.73%, respectively. The opposite variation of cellulose/hemicellulose and lignin might be caused by their respective chemical properties. Cellulose and hemicellulose were amenable to biodegradation, while lignin was recalcitrant to the AD process. Thus, the digestion time for cellulose and hemicellulose was much shorter than that of lignin, e.g., 14, 2 and 60 days, respectively [40].

The cellulose removal rates of MA0, MA0.2 and MA1.6 were 62.46% ± 2.30%, 68.23%± 2.01% and 69.90% ± 3.12%, respectively, and the hemicellulose removal rates of MA0, MA0.2 and MA1.6 were 70.28% ± 1.40%, 73.10% ± 2.01% and 76.95% ± 3.32%, respectively (Fig. 6**b**). However, the lignin removal rates of MA0, MA0.2 and MA1.6 were much lower, reaching 16.48% ± 2.71%, 26.85% ± 1.20% and 30.58% ± 2.17%, respectively. Such data could indicate that micro-aeration increased the degradation rates of cellulose, hemicellulose and lignin to different extents, which was consistent with previous studies [10,11,41]. In addition, higher micro-aeration intensity led to higher cellulose, hemicellulose and lignin degradation, e.g., the cellulose, hemicellulose and lignin removal rates of MA1.6 are 12.05%, 9.53% and 85.51% higher than those of MA0. The higher degradation of corn straw and lower methane yield in MA1.6 could be explained by the consumption of available substrate by facultative bacteria using oxygen as an electron acceptor and the inhibition of methanogenesis by excess oxygen.

3.6. Microbial community structure analysis

The digestate samples from MA0, MA0.2 and MA1.6 used for the microbial community analysis were collected at the 13th day of AD. The Shannon diversity indexes of MA0, MA0.2 and MA1.6 were 3.26, 3.78 and 4.30, respectively, which reflected higher microbial diversities under micro-aeration condition (**Fig. S4**). Similarly, Lim et al. [42] and Fu et al. [35] discovered that microbial diversity was positively affected by the oxygen. As shown in Fig. 7**a**, the top 4 dominant phyla for MA0 and MA0.2 were *Bacteroidota*, *Firmicutes*, *Synergistota* and *Spirochaetota* in that order, while the top 4 dominant phyla for MA1.6 were *Firmicutes*, *Bacteroidota*, *Synergistota* and *Spirochaetota* in that order. *Bacteroidota* was reported to possess the ability to decompose cellulose into cellobiose and glucose and *Firmicutes* could secrete extracellular hydrolytic enzymes (e.g., cellulase, lipase and protease), thus accelerating the degradation of large molecules such as cellulose, protein and lipids [42,43]. The increased relative abundance of *Firmicutes* from MA0 to MA1.6 indicated the improved abilities of extracellular enzymes production under micro-aeration conditions. *Synergistota*, also named *Synergistetes*, can produce short chain fatty acids and act as syntrophic acetate oxidizers [44]. *Spirochaetota* was a kind of acidogenic bacterium that could convert carbohydrates into simple VFAs [45]. The relative abundance of *Spirochaetota* of MA0, MA0.2 and MA1.6 accounted for 1.20%, 2.80% and 5.60%, respectively, which was in agreement with the reported results, i.e., 0.14–5.91% in the AD system with corn straw as the substrate [46]. The substantial increase of *Spirochaetota* from MA0 to MA1.6 suggested the promotion of the acidification process by micro-aeration. The relative abundance of *Firmicutes*, *Synergistota and*

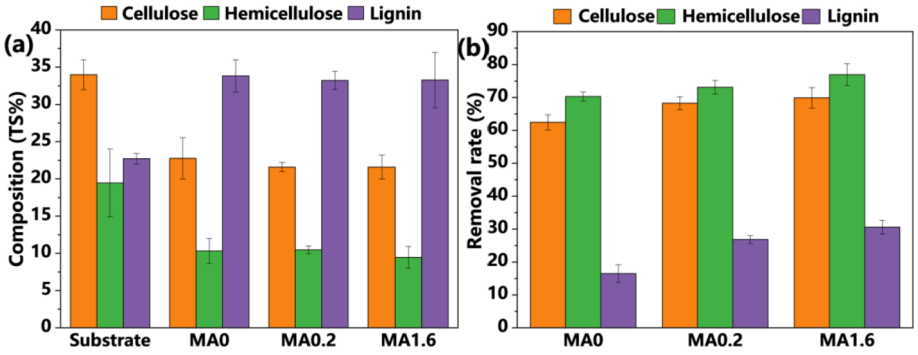


Fig. 6. Structural composition of raw corn straw and biogas residues (a) and removal rate of different components (b) during AD.

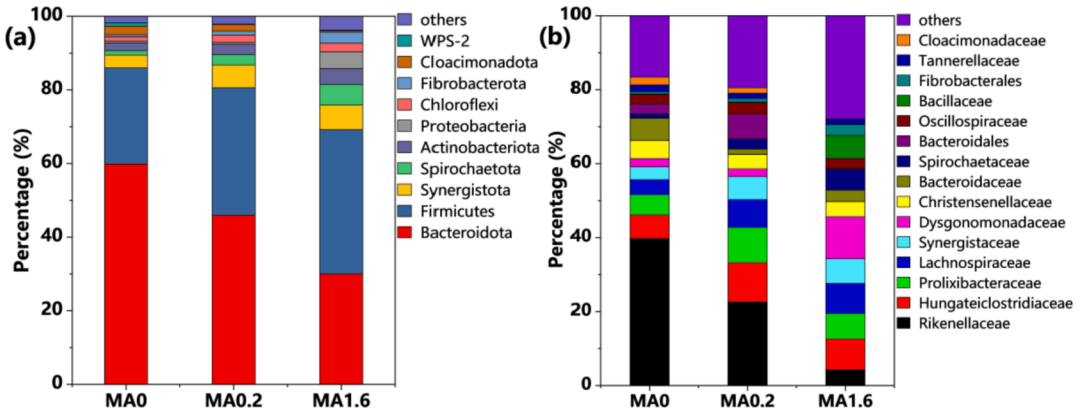


Fig. 7. Bacterial community structures of MA0, MA0.2 and MA1.6 at phylum (a) and family (b) levels. Phylum or family less than 1% of the total sequences and unclassified phylum or family were classified as others.

Spirochaetota of MA0.2 were 32.56%, 77.14% and 133.33% higher than that of MA0, respectively. The relative abundance of these three phyla of MA1.6 were 50.19%, 91.42% and 366.67% higher than that of MA0, respectively. These findings indicated the enhanced ability of AD to metabolize complex organics into small and simple molecules under micro-aeration conditions.

Bacterial community structures of MA0, MA0.2 and MA1.6 at family level are shown in Fig. 7**b**. The relative abundance of *Lachnospiraceae* affiliating with phylum *Firmicutes* was 7.49% in MA0.2 and 8.14% in MA1.6, which were 117.68% and 181.32% higher than that in MA0, respectively. *Lachnospiraceae* was responsible for the degradation of lignocellulose with high hydrolysis efficiency [47,48]. The increasing relative abundance of family *Lachnospiraceae* from MA0 to MA1.6 demonstrated the increasing ability to break large cellulosic molecules of substrates into fermentable sugars under micro-aeration conditions. *Spirochaetaceae* (affiliating with phylum *Spirochaetota*) and *Synergistaceae* (affiliating with phylum *Synergistota*) were two well-known acetate-oxidizing bacteria, which were capable of oxidizing acetate into H_2 and CO_2 [49]. The relative abundance of *Spirochaetaceae* of MA0.2 and MA1.6 were 2.76% and 5.54%, which were 188.85% and 420.21% higher than that in MA0, respectively. Similarly, the relative abundance of *Synergistaceae* of MA0.2 and MA1.6 were 6.24% and 6.71%, which were 147.10% and 174.40% higher than that in MA0, respectively. Under micro-aerobic condition, the increase of the relative abundance of fermentative bacteria with the increase oxygen dose reflected the higher ability of AD to utilize the hydrolysis products from cellulosic biomass for producing VFAs.

The hydrolysis of cellulosic substrate, such as corn straw has been recognized as the rate-limiting step during AD [50]. Therefore, compared to MA0, the increase of the relative abundances of the hydrolysis and fermentative bacteria in MA0.2 and MA1.6 under micro-aeration condition could in turn lead to higher hydrolysis and fermentation rates.

The Shannon diversity indexes of archaea increased slightly in MA0.2, but decreased rapidly in MA1.6, compared with MA0 (**Fig. S4**), which indicated that the archaeal diversity was negatively affected by oxygen. The reason behind this result might be that archaea especially methanogens are obligate anaerobic microbiota and the occurrence of oxygen could inhibit the growth of archaeal community in AD. As shown in Fig. 8, the dominant methanogens at genus level in MA0, MA0.2 and MA1.6 were *Methanobacterium*, *Methanosarcina* and *Methanosaeta*. However, the OTU reads and relative abundance of the genera varied with the increase of oxygen. According to Fig. 8**a**, the OTU reads of *Methanobacterium*, *Methanosarcina* and *Methanosaeta* of MA0.2 were 1.32, 1.42 and 6.76 times of those of MA0 and the OTU reads of *Methanobacterium*, *Methanosarcina* and *Methanosaeta* of MA1.6 were 1.46, 0.12 and 9.06 times of those of MA0. *Methanobacterium* and *Methanosarcina* were dominant genera in both MA0 and MA0.2, while the dominant genera were only *Methanobacterium* (the relative abundance was 84.05% and 1.73 times of that of MA0) of MA1.6. All the three dominant archaeal genera can survive and even grow better in the presence of limited amount of oxygen. Nevertheless, the OTU reads of *Methanobacterium* and *Methanosaeta* continued to show better adaption even under high oxygen concentration (i.e., MA1.6), whereas the OTU reads of *Methanosarcina* decreased rapidly. The above-mentioned results indicated that the archaeal community compositions changed with micro-aerobic condition and the oxygen tolerance of *Methanosarcina* was lower than that of *Methanobacterium* and *Methanosaeta*.

4. Discussion

Using micro-aeration to regulate the microbial community has

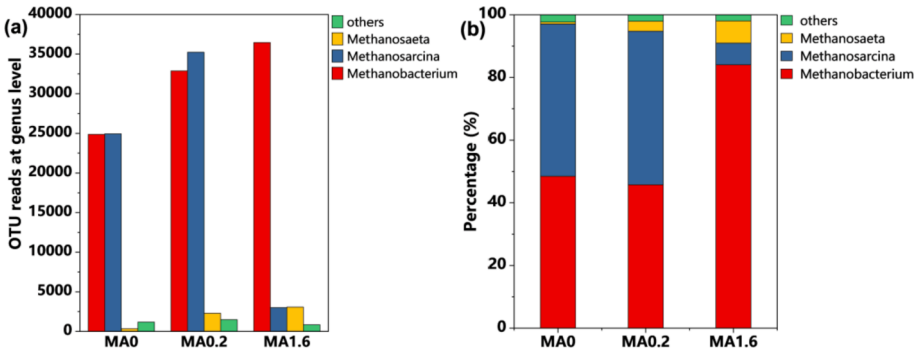


Fig. 8. Operational taxonomic unit (OTU) reads (a) and community structures (b) of archaea of MA0, MA0.2 and MA1.6 at genus level. Genus less than 1% of the total sequences and unclassified genus were classified as others.

attracted increasing attention due to its potential economic and environmental benefits. However, the underlying mechanisms of how micro-aeration benefit the AD process are largely unrevealed. In current study, the effects of micro-aeration on AD of corn straw were investigated. In order to figure out how micro-aeration benefits the AD process, comprehensively analytic methods including intermediate products analysis, model-based analysis, microbial community structure analysis, cyclic voltammetry (CV) and conductivity tests were conducted. Based on these experimental results, the mechanisms behind the benefits of micro-aeration were proposed and illustrated in Fig. 9. The oxygen introduced into the AD reactor could stimulate the growth of facultative microbes (e.g., phylum *Firmicutes* in Fig. 7) that in turn produced more extracellular enzymes (e.g., cellulase) [47,48] and enhanced the hydrolysis of corn straw. This inference could be confirmed by the higher VFAs and SHAs concentrations and higher removal rate of polymeric components of corn straw (i.e., cellulose, hemicellulose and lignin). Furthermore, *Spirochaetaceae* (phylum *Spirochaetota* in Fig. 7**b**) and *Synergistaceae* (phylum *Synergistota* in Fig. 7**b**) were equipped with the ability to oxidize acetate into H_2 and CO_2 and provide substrates (i.e., H_2 and CO_2) for methanogenesis [49], which were also stimulated by micro-aeration. The hydrogenotrophic methanogenesis was proposed as a preferable methane production pathway in AD reactors operated in micro-aerobic conditions [11]. The increasing VFAs concentrations (Fig. 3**A)** and relative abundance of *Synergistota* and *Spirochaetota* (acetate-oxidizing bacteria) (Fig. 7) under micro-aerobic condition suggested more substrate CO_2 and H_2 for the faster metabolisms of hydrogenotrophic methanogens.

In addition to the variation of bacterial communities, some archaea also acclimatized themselves to the micro-aeration condition. *Methanobacterium*, *Methanosaeta* and *Methanosarcina* were reported to be able to survive even grow better in the presence of a small amount of oxygen [41]. Generally, these three archaea belong to hydrogenotrophic methanogens, acetoclastic methanogens and versatile methanogens (capable of utilizing acetate, methylamines, methanol and H_2 /CO_2), respectively [51–53]. The archaeal diversities decreased in the presence of excess oxygen, while the oxygen-tolerant methanogens (e.g., *Methanobacterium*) became dominant and ensured the methanogenesis to produce methane.

Apart from the microbial community variations, here we firstly found the enhanced DIET based syntrophic methane production under micro-aeration could be another non-negligible factor. Microbial metabolism could convert some corn straw components into SHAs, which could serve as electron shutters and act as mediators for DIET [54–56]. The final concentrations of SHAs of MA0, MA0.2 and MA1.6 were 1.31 ± 0.06 g/L, 1.67 ± 0.12 g/L and 2.05 ± 0.10 g/L, respectively **(**Fig. 4**)**. The higher SHAs under micro-aerobic condition appeared to empower the AD system with higher capacity of electron transfer (proved by the FTIR analysis results, **Fig. S2** and CV experiment, Fig. 5**a**) and better conductive environment (proved by conductivity testing results, Fig. 5**b**) for syntrophic microorganisms, which indicated their stronger ability of DIET. Meanwhile, the relative abundances of organics-oxidizing bacteria, acetate-oxidizing bacteria ((e.g., *Spirochaetaceae* and *Synergistaceae* (Fig. 7)) and methanogenic archaea ((e.g., *Methanobacterium*, *Methanosarcina* and *Methanosaeta* (Fig. 8)) that associate with DIET also increased, which reflected the enhanced DIET under micro-aeration condition.

These aforementioned factors were considered to work together to improve methane yield under micro-aeration condition. However, the micro-aeration intensity over 0.2 mL/(g·VS·day) decreased cumulative methane yield (Fig. 1). Such a decrease under excess micro-aeration intensity could be explained by: 1) The excess nonproductive consumption of available substrates. The introduced oxygen could stimulate the growth of facultative microbes that could consume part of the substrates without producing methane, i.e., the substrates towards methane production were reduced; and 2) The inhibition of methanogens by oxygen. Although some methanogens, such as *Methanobacterium*, *Methanosarcina* and *Methanosaeta* have a certain of oxygen tolerance, their tolerance is limited. Therefore, overdose of oxygen during micro-aeration could result in an inhibition to these methanogens. Similarly, Yu et al. [13] also found a reduction of methane production in the AD reactor at high oxygen dose (50 and 150 mg O_2/L-reactor/cycle) when studying the effects of micro-aeration on the performance of AD of black-water.

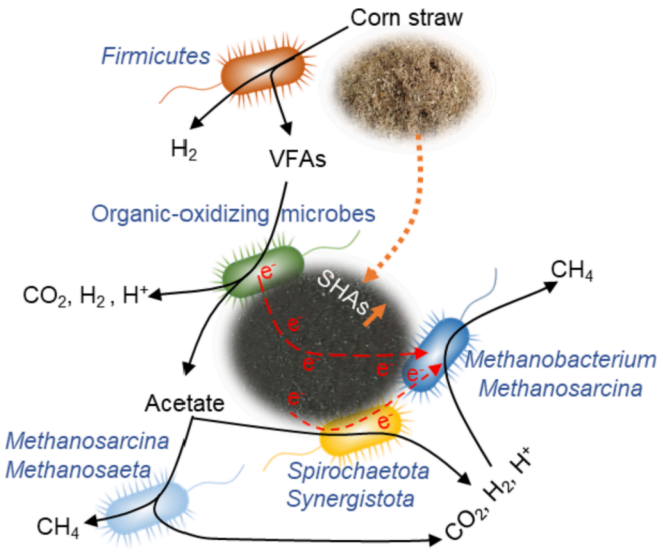


Fig. 9. Proposed mechanisms of micro-aeration to enhance the methane production of AD of corn straw. The microbes shown in the pathway remain increase in their relative abundance under micro-aeration.

5. Conclusions

This study evaluated the effects of micro-aeration on AD of corn straw. Micro-aeration could be conducive to the AD process depending on the oxygen dosage. Appropriate micro-aeration shortened the lag-phase and enhanced the methane yield, while excessive micro-aeration reduced the methane yield. Under proper micro-aeration, the relative abundances of facultative bacteria and oxygen-tolerant methanogens were increased, meanwhile the DIET during AD was also enhanced, thus improving the methane yield. However, excess micro-aeration increased the nonproductive consumption of available substrates (originally for methane production) while resulting in the inhibition of methanogens, which lowered the methane yield. This study shows micro-aeration is a promising approach to improve the methane production of AD. However, the micro-aeration intensity should be well controlled.

CRediT authorship contribution statement

Rong Zhu: Investigation, Methodology, Data curation, Writing – original draft. **Dong-Hui Wang:** Validation, Data curation. **Yi Zheng:** Writing – review & editing. **Hua Zou:** Resources, Supervision, Visualization. **Shan-Fei Fu:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fuel.2022.123604.

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